Roots and Powers

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 69 minutes · Total marks: 69

KEY RULES

Squares and cubes	1^2 to 15^2; $2^3, 3^3, 4^3, 5^3, 10^3$
Roots	$\sqrt{25} = 5$; $\sqrt[3]{27} = 3$
Reciprocal	The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$
Powers	$a^2 = a \times a$; $a^3 = a \times a \times a$

WORKED EXAMPLE 1

[2 marks]

Work out $3^3 - \sqrt{49}$

Step 1 Powers and roots $27 - 7$

Answer: 20

WORKED EXAMPLE 2

[2 marks]

Write 2^{-3} **as a fraction.**

Step 1 A negative power means the reciprocal $\frac{1}{2^3} = \frac{1}{8}$

Answer: $\frac{1}{8}$

WORKED EXAMPLE 3

[3 marks]

Work out $\left(\frac{4}{9}\right)^{\frac{1}{2}}$

Step 1 The power $\frac{1}{2}$ means square root $\frac{\sqrt{4}}{\sqrt{9}}$

Step 2 Work out each root $\frac{2}{3}$

Answer: $\frac{2}{3}$

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1 7 $\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$

Find the value of n

$n =$ .....

(Total for Question 7 is 2 marks)

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2 7 (a) Simplify $(4^{-2})^0$

..... (1)

$3^{-14} \times 3^8 = 3^m$

(b) Find the value of m

$m =$ ..... (1)

(Total for Question 7 is 2 marks)

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3 8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$n =$ .....

(Total for Question 8 is 2 marks)

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4 $\frac{y^{\square} \times y}{y} = y^6$

Find the value of n.

$n =$ .....

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5 Given that $150^x = 1$

(a) write down the value of x.

$x =$ .....

(1)

Given that $3^{-8} \div 3^{-6} = 3^n$

(b) find the value of n.

$n =$ .....

(1)

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6 **12** Solve $2^{-4x} = 32$

$x =$ .....

(Total for Question 12 is 2 marks)

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12 $3^{\frac{1}{2}} \times 3^{\frac{2}{5}} = 3^m$

(a) Work out the value of m

$m =$ .....

(1)

$^{-10} \div 5^{-4} = 5^n$

b) Work out the value of n

$n =$ .....

(1)

(Total for Question 12 is 2 marks)

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8

Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

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9 (a) Write $2^3 \times 2^4$ as a single power of 2

.....

(1)

(b) $280 = 2^n \times 5 \times 7$

Find the value of n.

$n =$ .....

(2)

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10 Given that $\frac{3^x}{9^{3x}} = 81$

find the value of x.
Show clear algebraic working.

$x =$ .....

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11 **14** Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

$n =$ .....

(Total for Question 14 is 3 marks)

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12 $128 = 4^{2x} \times 2^x$

Work out the value of x.

$x =$ .....

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13 (a) Write down the value of m, given that $3^4 \times 3^5 = 3^m$

$m =$ .....

(1)

(b) Write down the value of n, given that $(5^3)^7 = 5^n$

$n =$ .....

(1)

(c) Find the value of p, given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p =$ .....

(2)

(Total for Question 1 is 4 marks)

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14 (a) Write $2^3 \times 2^6$ as a single power of 2

.....

(1)

(b) Write $\frac{3^9}{3^4}$ as a single power of 3

.....

(1)

(c) $\frac{5^n}{5^4 \times 5^6} = 5^3$

Find the value of n.

$n =$ .....

(2)

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15 **6** (a) Write down the value of x^0

.....

(1)

Given that $2^{-3} \times 2^9 = 2^n$

(b) find the value of n

$n =$ .....

(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

(c) find the value of m

$m =$ .....

(2)

(Total for Question 6 is 4 marks)

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16 $\dfrac{8}{2^7} = 2^n$

(a) Find the value of n.

$n =$ .....

(2)

$(13^{-6})^4 \times 13^5 = 13^k$

(b) Find the value of k.

$k =$ .....

(2)

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CHALLENGE QUESTIONS

17 11 $\sqrt{2} \times 16 = 2^x$

(a) Find the value of x.
Show your working clearly.

$x =$ .....

(2)

$$\frac{11^{-6})^5}{11^4} = 11^n$$

b) Find the value of n.
Show your working clearly.

$n =$ .....

(2)

(Total for Question 11 is 4 marks)

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18 Simplify $\dfrac{30 \times 25^{2x+1}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

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19 Given that $M = \dfrac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

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20 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

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21 (a) Simplify $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form 2^a where a is an integer.

Show each stage of your working clearly.

.....

(3)

Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$

(b) find the value of n.

$n =$ .....

(4)

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TOTAL FOR PAPER IS 69 MARKS